INCH-POUND

MIL-C-48087A (AR)
AMENDMENT 6
7 December 1992
SUPERSEDING
AMENDMENT 5
8 September 1988

MILITARY SPECIFICATION

CARTRIDGE, 4.2 INCH, HE, M329A2
W/SUPPLEMENTARY CHARGE
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-C-48087A (AR), dated 17 June 1982, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2, Under "PRODUCT AND PACKAGING DRAWINGS"

Add "12624505 - Packing and Marking for box, Fiberboard for Obturator Assembly"

* Add "9244177 - Charge, Propellant: M36A2"

* Delete "12624505 - ...." in its entirety and substitute the following:

"12589951 - Packing and Marking for Box, Fiberboard for Obturator or Obturator assembly
12598564 - Packing and Marking for Box, Wood, Wirebound Pallet Type for Obturator or Obturator Assembly"

2.1.2, Under "INSPECTION EQUIPMENT DRAWINGS"

* Add "12910223 - Obturator Support Gage"

AMSC N/A FSC 1315

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PAGE 5

* 3.8, Add new paragraph as follows:

"3.8 Propellant charge correctness. The propellant charge shall be in accordance with drawing 9244177. There shall be a total of five (5) five-increment bundles, eight (8) single increments, two (2) half increments and one (1) bag loading assembly.

PAGE 7

Table II:

Delete in its entirety and substitute the new paragraph, PAGE 7, included.

* Delete "Obturator/Examination for Defects ... 25 ... 100% ... 3.2 ... 4.4.2.4" and substitute the following:

"Obturator/Examinations for Defects ... 25/Note A ... 100% ... 3.2 ... 4.4.2.4"

* Add under NOTES, "Note A: Twenty-five samples per cavity per mold

PAGE 12

4.4.2.4, Delete in its entirety and substitute the new paragraph, PAGE 12, included.

* 4.4.2.4

* Major 101, delete in its entirety and substitute the following:

"Diameter of bristle strip well ... 0.40% ... 3.2 ... Gage/ 12910223 (Two measurements, 90 degrees apart) Note 1"

* Add under NOTES, "Note 1: Diameter of strip bristle well is equal to sum of 3.775 inches plus two opposite dial indicators readings"

* Major 106, delete in its entirety and substitute the following:

"Hardness ... 0.40% ... 3.2 ... Gage"

* Major 111, delete in its entirety

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TABLE II First article inspection

MIL-C-48087A (AR)
AMENDMENT 6

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge Assembly				
	Examination for defects	80	100%	3.2	4.4.2.8
	Proving ground	80	100%	3.5	4.4.3.1/4.5.2
	Obturator				
	Examination for defects	25	100%	3.2	4.4.2.4
	Obturator Assembly				
	Examination for defects	50	100%	3.2	4.4.2.14
	Bristle, Strip				
	Examination for defects	25	100%	3.2	4.4.2.15

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.4	Obturator		9391032
			NEXT HIGHER ASSEMBLY 9391030

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of strip bristle well		0.40%	3.2	Gage
102	Width of strip bristle well		0.40%	3.2	Gage/Caliber
103	Thickness at front face (Large) (Four measurements, 90 degrees apart)		0.40%	3.2	Gage/Caliber
104	Thickness at rear face (Small) (Four measurements, 90 degrees apart)		0.40%	3.2	Gage
105	Distance from front face to strip bristle well		0.40%	3.2	Gage/Caliber
106	Separator cut(s) missing		0.40%	3.2	Visual
107	Forward inside diameter		0.40%	3.2	Gage
108	Overall width of obturator		0.40%	3.2	Gage
109	Depth of separator cut		0.40%	3.2	Gage
110	Concentricity of strip bristle well		0.40%	3.2	Gage
111	Evidence of contaminants (foreign materials) in bristle strip well considered capable of interfacing with adhesion		0.40%	3.2	Visual
Minor					
201	Angle at rear obturator		0.65%	3.2	Visual
202	Radii missing		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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* 4.4.2.8

Delete Critical defect 3 in its entirety and add the following:

"Special a, Metal parts separation (proving ground) ... 80/50 ... 3.5.6 ... 4.4.3.1.2/4.5.2"

Major 105, delete in its entirety and substitute the following:

"Obturator assembly improperly assembled ... 100% ... 3.2 ... Gage"

Major 106, change "obturator" to "obturator assembly"

* Add new Major 112 as follows:

"Propelling charges incorrect ... 200-0-1 ... 3.8 ... Visual"

PAGE 22

4.4.2.14, Delete in its entirety and substitute the new paragraph, Page 22A and 22B, included.

4.4.2.15, Delete in its entirety and substitute the new paragraph Page 22C, included.

4.4.2.16, Add new paragraph, Page 22D, included.

4.4.2.16, Delete "12624505" and substitute "12589951"

4.4.2.17, Add new paragraph, Page 22E, included.

PAGE 22A

* 4.4.2.14, Delete Major 105 in its entirety and substitute the following:

"Backing material of bristle strip exceeded .022 inch max. above the flush obturator surface."

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.14	Obturator Assembly		9391030
			NEXT HIGHER ASSEMBLY 9235654

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Improper orientation of bristle strips when viewing the obturator assembly from rear edge		100%	3.2	Gage
b.	Obturator assembly outside diameter		100%	3.2	Gage/Note 1
Major					
101	Location and depth of separator cuts		0.40%	3.2	Gage
102	Location of staples		0.40%	3.2	Gage
103	Paint missing or rear edge of obturator assembly		0.40%	3.2	Visual
104	Clinched ends of staple shall be below flush with the rubber surface of inside diameter of obturator		0.40%	3.2	Visual
105	Backing material of bristle strip exceeded obturator surface		0140%	3.2	Gage/Note 2

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.14	Obturator Assembly (continued)				9391030
					NEXT HIGHER ASSEMBLY
					9235654
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major(cont'd)					
106	Crown of each four (4) fastening staples exceeded 0.018 inch max above flush of obturator surface		0.40%	3.2	Gage/Note 2
107	Staple broken or missing		0.40%	3.2	Visual
108	Leg of staple not within the backing material of bristle strip		0.40%	3.2	Visual
109	Bristle strip ends not stapled within obturator groove		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

1. Gaging for outside diameter shall be performed with obturator assembled onto a tapered plug with a diameter of 3.775 inches.
2. Gaging shall be performed with obturator assembled onto a tapered plug with a diameter of 3.769 inches.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9391031
4.4.2.15	Bristle, Strip				NEXT HIGHER ASSEMBLY 9391030
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width of strip		0.40%	3.2	Gage
102	Length of strip		0.40%	3.2	Gage
103	Height of wire bristles, max (3 places)		0.40%	3.2	Gage
104	Height from the bottom wire bristles to the top backing (3 places)		0.40%	3.2	Gage
105	Angle of wire bristles		0.40%	3.2	Gage/Visual Note 1
106	Distance from the end of bristle strip to the first row of wire bristles		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

1. Fifteen wire bristles in one linear inch of the bristle strip shall be inspected, if four or more wire bristles fail to comply with the applicable requirement, the bristle strip shall be classed as defective.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48087A (AR)
AMENDMENT 6

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Packing and Marking for Box, Fiberboard for Obturator Assembly	1 OF 1	12624505
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Lot number missing		0.65%	5.1/5.2	Visual
202	National stock number missing		0.65%	5.1/5.2	Visual
203	Item nomenclature/part number missing or incorrect		0.65%	5.1/5.2	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.2.17, Delete in its entirety.

* 4.4.3.1, Change "4.5.1" to "4.5.2"

* 4.4.3.1.2.1, Delete in its entirety and substitute the following:

"4.4.3.1.2.1 Beginning with the first lot produced and continue until three consecutive lots have been found to be acceptable without the exclusion of an outlier (see paragraph 4.4.3.1.3.1), eighty (80) sample cartridges shall be randomly selected from each lot."

* 4.4.3.1.2.2, Delete in its entirety and substitute the following:

"4.4.3.1.2.2 After three consecutive lots have compiled with the criteria of paragraph 4.4.3.1.2.1, fifty (50) sample cartridges shall be randomly selected from each lot. In the event that a lot fails to comply with the requirements (TABLE III), or an outlier must be excluded to meet the range probable error requirement (see paragraph 4.4.3.1.3.2), the sampling criteria shall revert to as specified in paragraph 4.4.3.1.2.1.

TABLE III

		Acceptance Criteria			
		Sample Size 80		Sample Size 50	
Category	Defect	Accept	Reject	Accept	Reject
Critical	Short Round (less than 80% mean range, para. 3.5.3)	0	1	0	1
Critical	Premature burst	0	1	0	1
Special	Metal parts separation	0	1	0	1
Major	Misfire	1	2	1	2
Major	Dud & Outlier (Combined)"	3	4	2	3

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* 4.4.3.1.3.1, Add new paragraph as follows:

"4.4.3.1.3.1 Range probable error calculation (Increment 34, forty (40) samples). For a sample size of forty (40), two (2) outliers are permitted to be excluded in the calculation of range probable error. If the range probable error requirement (paragraph 3.5.4.2) is not met after the exclusion of the two shortest outliers, the lot shall be rejected."

PAGE 23

* 4.4.3.1.3.2, Add new paragraph as follows:

"4.4.3.1.3.2 Range probable error calculation (Increment 34, twenty-five (25) samples). For a sample size of twenty-five (25), one outlier (paragraph 6.5.1) is permitted to be excluded in the calculation of the range probable error. The following applies if the range probable error requirement (3.5.4.2) is not met after the exclusion of the shortest outlier. The lot shall be rejected if three (3) or more outliers are experienced. The lot shall be retested utilizing another weapon if two (2) outliers are experienced. Forty (40) additional test samples shall be randomly selected and fired at 34 increments in the retest. A new range probable error shall be calculated of the based only on the forty (40) additional test samples. One outlier is permitted to be excluded in the new range probable error. If two (2) or more outliers occurred in the retest, the lot shall be rejected."

PAGE 26

5.1, Delete in its entirety and substitute the following:

"5.1 Packaging requirements.

a. Packaging for 4.2 Inch Mortar Ammunition shall be as specified on drawing 9272389

b. Packaging for obturator and obturator assembly shall be as specified on drawing 12589951."

5.2, Delete in its entirety and substitute the following:

"5.2 Packing and marking.

a. Packing and marking for 4.2 Inch Mortar Ammunition shall be as specified on drawing 9243973.

b. Packing and marking for obturator and obturator assembly shall be as specified on drawing 12598564."

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* 6.5.1, Add new paragraph as follows:

"6.5.1 Outlier criteria for increment 34. A round shall be classified as a statistical outlier provided that it is not a critical short round (less than 80% of the mean range calculated without that round) and it meets the conditions of 6.5.1.1 or 6.5.1.2.

* 6.5.1.1, Add new paragraph as follows:

"6.5.1.1 Single outlier. An out-of-pattern round (with the lowest range value of the group) shall be classified as an outlier if it meets the following:

$$(\overline{X} - X_{out})/s' > Tn$$

Where $\overline{X}$ = Arithmetic average range for all test samples
X_{out} = Range of the out-of-pattern round
s' = Standard deviation for all test samples
Tn = 3.009 (for sample size of 25)
= 3.240 (for sample size of 40)

* 6.5.1.2, Add new paragraph as follows:

"6.5.1.2 Multiple outlier. If the round with the lowest range value is determined to be an outlier in accordance with 6.5.1.1 and the range probable error requirement is not met after exclusion of that round, then round with the next lowest value shall be classified as an outlier if it meets the following:

$$(\overline{X} - X_{out})/s' > Tn$$

Where $\overline{X}$ = Arithmetic average range for all test samples
X_{out} = Range of the out-of-pattern round
s' = Standard deviation for all test samples
Tn = 2.987 (for sample size of 24)
= 3.228 (for sample size of 39)

Similarly, a third outlier is determined the same way with the exception that the two outliers are excluded in the X and s' calculation, and Tn = 2.963 (n=23).

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* 6.10, Add new paragraph as follows:

"6.10 Process aid for Comp-B loading. Only qualified process aid materials are permitted to be used in Composition B. The following materials have been found to be acceptable wax dispersants when added to molten Composition B:

Lecithin, MIL-L-3061 in quantity of 0.1% - 0.02% by weight added.

Pegosperse 400 DS in quantity of 0.08% - 0.02% by weight added.

The Composition B temperature shall be above 194 degrees Fahrenheit to provide acceptable cast results.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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